

Niteesh Kanungo

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PROFESSIONAL SUMMARY

- Data Science Professional with 4+ years of experience in the Research Industry and in Academia with a strong background in **Machine Learning, Data visualization, Statistics and Natural Language Processing**.
- Strong familiarity and experience with the end-to-end pipeline of data analysis viz. **data pre-processing, mining and visualization** for the purposes of **Classification, Regression and Deep Learning** using Python, R and SAS.
- Experience with Machine Learning Algorithms such as - **Linear regression, Logistic regression, Discriminant Analysis, Tree-based methods, PCA, t-SNE, Clustering, SVM, Neural Networks – DNN, CNN, LSTM**.
- Good understanding of Natural Language Processing: **Tagging, Data parsing, Topic modeling, Sentiment Analysis**, and Application of deep learning to NLP.
- Experience with building **Computer vision, image recognition, classification models** using OpenCV.
- Certified and having extensive knowledge of Cloud technology: **Google Cloud Platform and AWS**.
- Excellent working knowledge in **UNIX and Linux shell** environments using command line utilities.
- Cognitive about designing, deploying and operating highly available, scalable and fault-tolerant systems using **Amazon Web Services (AWS)**.
- Experience with **A/B Testing, cohort analysis and churn rate prediction**.
- Experience working with notebook technologies like **Jupyter, Apache Zeppelin** and open Source distribution **Anaconda**.
- Worked on various applications using python integrated IDEs like **Eclipse, IntelliJ, Sublime Text and PyCharm**.
- Experience of using Python visualization libraries like **Pandas, Matplotlib, SciPy, Seaborn and Go**.
- Experience in project deployment using **Heroku, Jenkins and Amazon Web Services (AWS)**.
- Experience with version control systems like **Git and Enterprise GitHub**.
- Experience in working with GPS tracking/Streaming (IOT) devices, built a Geo-fencing module and **AWS IOT** for storing and querying data.
- Working knowledge of Financial Analysis (**Time series Data**) and **Algorithmic Trading** using Quandl.
- Good experience of software development in Python (libraries used: libraries- **Beautiful Soup, NumPy, SciPy, Matplotlib, Pandas data frame, network, urllib2, MySQL** for database connectivity)
- Experience with **Jenkins**. Worked on Jenkins continuous integration tool for deployment of project.
- Experience with Streaming platforms like **Apache Kafka and GCP PubSub**.
- Working knowledge on **Kubernetes** to deploy scale, load balance, and manage **Docker containers** with multiple namespace versions.
- Knowledge in **Data mining and Data warehousing** using ETL Tools and Proficient in Building reports.
- Experience of working with relational databases like **Oracle, SQLite, PostgreSQL, MySQL** and non-relational databases like **HBase and Cloud Bigtable**.
- Experienced with **Unit Testing, System Integration Testing (SIT)** and **User Acceptance Testing (UAT)**.
- Strong understanding of Big data technologies including **Apache Spark (Pyspark), Hadoop, HDFS, Hive, Pig and MLlib**.
- Good Experience in building Dashboards using **Google and Tableau Analytics**.
- Experience dealing with various development methodologies like **Waterfall, Scrum, and Agile**.
- Excellent Interpersonal and communication skills, efficient time management and organization skills, ability to handle multiple tasks and work well in a team environment.
- Experienced in complete **Software Development Life Cycle**, and major aspects of technological projects including **Requirement Analysis, Design, Development, Database Design, Deployment, Testing, Debugging, and Application Maintenance**.

EDUCATION

University of Illinois at Urbana-Champaign. Illinois, USA	Dec. 18
• Master's (M.S.) in Information Management. (Teaching Assistant: CS598)	
VIT University. Vellore, TN, India.	Aug. 14
• Master's (M.S.) in Bio-informatics and Biotechnology. (Teaching Assistant: BST504, BMG502)	
Christian Eminent College. Indore, MP, India	July 12
• Post Graduate Diploma in Computer Applications (PGDCA)	
Institute of Professional Studies (DAVV). Indore MP, India	Aug. 11
• Bachelor's (B.S.) in Computer Science.	

TECHNICAL EXPERTISE

Operating System	Unix, Linux-Ubuntu, Kali, CentOS, Windows and MacOS
Programming Languages	Python, JavaScript, Java, Octave, Scala (Inter), R(Inter), Shell Scripting
Database	MySQL and PostgreSQL, HDFS
Analytical Tools	Numpy, Pandas, SciPy, Matplotlib, HiveQL, Tableau.
Cloud Technologies	Amazon Web Services, Google Cloud Platform, Databricks
Big Data Technology	Apache Hadoop, Apache Spark, Apache Sqoop
Machine Learning Tools	TensorFlow, Scikit Learning, MLlib, Keras and Weka
Deployment Tools	Heroku, Jenkins
Data Cleaning	Open Refine (Google)
Tools	Spyder, Visual Studio, Tableau Analytics
Defect Tracking	JIRA and VersionOne
Frameworks	Flask and Django
Version Control Systems	GIT, GITHUB
IDE's/Development Tools	PyCharm, IntelliJ, Atom, Eclipse and Sublime Text.

WORK EXPERIENCE

Data Scientist (Intern) - Bazaarvoice, Austin, Texas, USA

May 18 – Aug. 18

- Developed a Deep Learning (CNN) Model (82% acc.) using Keras, AWS Sagemaker for auto-moderation of images, saving on replacement with manual Moderation up to \$92k-\$100k in a year.
- Developed a script using NLTK, Flask-API running on AWS to get insights from thousands of reviews posted on client websites.
- Re-written an R-script for review analysis to Python to already integrated solutions within the network.
- Python automated scripting using Boto3 library for data quenching from AWS S3 buckets.
- Used Pandas and Matplotlib for Statistical Analysis of Salesforce data from Customer feedback.
- Used Docker images to restrict the workflow breaking from dependency updates.
- Used JIRA to assign, track the stories and tasks.
- Developed Restful Microservices using Flask and deployed single point of entry on AWS servers using EC2 instances.
- Automated most of the daily task using python scripting.

Programmer/Data Analyst - Abbott Power Plant, UIFS, Illinois, USA

June 17 – May 18

- Developing dashboards from Data sources generated by 10,000 data points at the plant.
- Estimating and analyzing the production cost as well as improving efficiency with predictive modeling. Also, calculating emissions with EPA norms.
- Analyzed various logs that are been generated and used various Python libraries to predict/forecast next occurrence of event with notification.
- Built a Monte Carlo simulation for predicting behavior of Data points and evaluating errors using pandas and Matplotlib.
- Used OpenRefine & Python to clean the data, created a relational model of the dataset using SQL on SQLite database. Generated prospective provenance information of the process involved using YesWorkflow and datalog.
- Built various graphs for business decision-making using Python matplotlib library.
- Designed and developed various analytical reports from data sources by blending data onto a single worksheet.
- Worked on web-scraping the Webpages and used modules like urllib2, Beautiful Soup and pandas.
- Automated the existing scripts for performance calculations using Numpy.
- Debugging and testing applications and fine tuning the performance

Business Intelligence Developer – Blazon Creations Ltd, Indore, MP, India

June 15 – Dec.16

- Improved operational efficiency of logistics by 7% of a major Retail company through Inventory Management and analysis of GIS Data.
- A/B testing to optimize churn rate and Improved Web traffic of client's website by up to 20% through Web Analytics, SEO and Web Portal optimizations.
- Composed moderate to complex SQL queries to analyze large and complex data sets.
- Wrote MapReduce code to make un-structured data into semi- structured data and loaded into Hive tables.
- Performed advanced procedures like text analytics and processing, using the in-memory computing capabilities of Spark.
- Played a key role in Supporting and deploying Cloud computing services, including IaaS, PaaS, SaaS deployments.
- Worked with application teams to install operating system, Hadoop updates, patches, version upgrades as required.

- Involved in analysis, specification, design, and implementation and testing phases of Software Development Life Cycle (SDLC) and used Agile methodology for developing application.
- Involved in creating Hive tables, loading with data and writing hive queries that will run internally in map reduce way.
- Making recommendations to the team in terms of appropriate testing techniques, shared testing tasks.
- Participate in requirement gathering and analysis phase of the project in documenting the business requirements by conducting workshops/meetings with various business users.

Computational Biologist – Eminent Biosciences (EMBS), Indore, MP, India

Sept.14 – May 15

- Statistical analysis of Microarray, Nextgen sequencing data using computational approach and open source tools.
- Developed scripts for automating tasks using Python and UNIX shell scripting.
- Worked on Python scripts to parse JSON documents and load the data in database.
- Maintained technical documentation for resolved issues for future reference
- Worked on Python OpenStack APIs and used NumPy for Numerical analysis.
- Setup automated cron jobs to upload data into database, generate graphs, bar charts, upload these charts to wiki, and backup the database.
- Ensured high quality data collection and maintaining the integrity of the Healthcare data.
- Resolving Complexity in the scripts of the website due to the complex logic and correlations.
- Designed and developed data management system using MySQL
- Wrote python modules to extract/load asset data from the MySQL source database.
- Used PyUnit, a python unit test framework for all Python applications.
- Ability to successfully implement the application in LINUX environment
- Carried out various mathematical operations for calculation purpose using Python library Numpy.

Graduate Research Fellow– Jawahar Lal Nehru University, New Delhi, India

Dec. 13 – Aug. 14

- X-Ray diffraction data preprocessed/analyzed using HKL2000 and XDS, collected at source (JNUCAR) and synchrotron (Indus-1,2 and RRCAT) for Drug Design, Protein structure prediction.
- Structure determination by experimental Phasing using CCP4 and Phenix suite, visualization and refinement using Coot (Linux).
- Responsible for handling the integration of database system.
- Developed rich user interface guidelines and standards throughout the development and maintenance of the website using CSS, HTML and JavaScript
- Upgraded existing UI with HTML, CSS, jQuery and Bootstrap.
- Involved in Design, Development, Deployment, Testing, and Implementation of the application.
- Designed Interface using Bootstrap framework.
- Coding and execution of scripts in Python/Unix/VB.
- Design, develop, test, deploy and maintain the website.
- Used UML Tools to develop Use Case diagrams, Class diagrams, Collaboration and Sequence Diagrams, State Diagrams and Data Modeling.

REFERENCES

*Will be provided as requested.